

3.2 Assignment: Using Data to Improve MLB Attendance

Author: Nicholas Stirling

Due Date: 12/21/25

Description: This assignment uses data on the Los Angeles Dodgers Major League Baseball (MLB) team. The data will be used to make a recommendation to management on how to improve attendance. Tell a story with the analysis and clearly explain the steps taken to arrive at that conclusion.

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("dodgers-2022.csv")

print(df.head())
df.info()
```

	month	day	attend	day_of_week	opponent	temp	skies	day_night	cap	shirt	\
0	APR	10	56000	Tuesday	Pirates	67	Clear	Day	NO	NO	
1	APR	11	29729	Wednesday	Pirates	58	Cloudy	Night	NO	NO	
2	APR	12	28328	Thursday	Pirates	57	Cloudy	Night	NO	NO	
3	APR	13	31601	Friday	Padres	54	Cloudy	Night	NO	NO	
4	APR	14	46549	Saturday	Padres	57	Cloudy	Night	NO	NO	

	fireworks	bobblehead
0	NO	NO
1	NO	NO
2	NO	NO
3	YES	NO
4	NO	NO

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 81 entries, 0 to 80
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   month           81 non-null    object
1   day             81 non-null    int64
2   attend          81 non-null    int64
3   day_of_week     81 non-null    object
4   opponent        81 non-null    object
5   temp            81 non-null    int64
6   skies           81 non-null    object
7   day_night       81 non-null    object
8   cap             81 non-null    object
9   shirt           81 non-null    object
10  fireworks       81 non-null    object
11  bobblehead      81 non-null    object
dtypes: int64(3), object(9)
memory usage: 7.7+ KB

```

Upon initial inspection of the loaded file, attendance is already a proper data type and there are no missing fields in any of the columns.

Initially, I will review the days of the week to see which days historically have the lower attendance. In order to do this I will calculate the average attendance per day of the week. I will compare these using a bar chart as that will make it easy to interpret and potential communication to management later on for the identified recommendation. I am expecting that weekday games are likely to have lower attendance then weekends.

```

In [3]: day_attendance = (df.groupby("day_of_week")["attend"].mean().sort_values())

print(day_attendance)

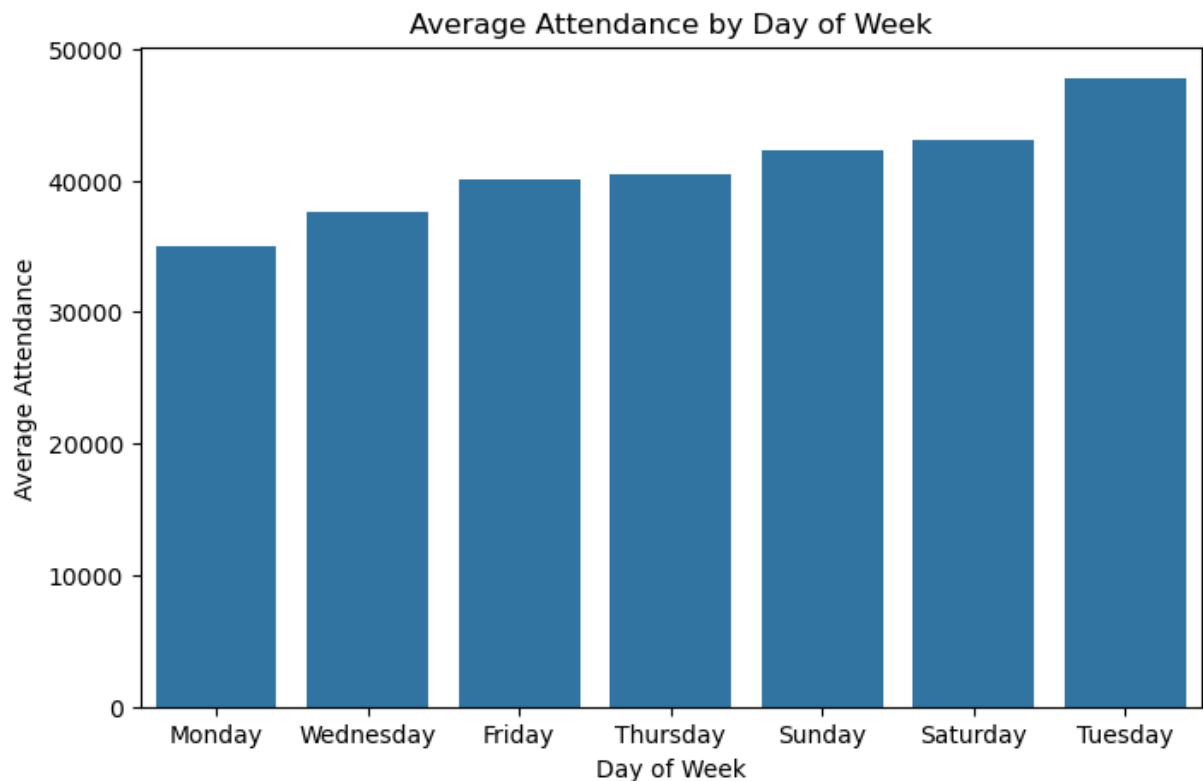
plt.figure(figsize=(8,5))
sns.barplot(
    x=day_attendance.index,
    y=day_attendance.values
)

plt.title("Average Attendance by Day of Week")
plt.xlabel("Day of Week")

```

```
plt.ylabel("Average Attendance")
plt.show()
```

```
day_of_week
Monday      34965.666667
Wednesday   37585.166667
Friday      40116.923077
Thursday    40407.400000
Sunday      42268.846154
Saturday    43072.923077
Tuesday     47741.230769
Name: attend, dtype: float64
```



Suprisingly Tuesday is actually the highest average attendance day for games, followed by the weekend games. Monday and Wednesday are lower on average by quite a margin and are the strongest candidates to be targeted for promotional strategies to boost attendance.

Next, I want to evaluate whether the promotional columns (assumed to be giveaways/entertainment) actually move the needle. I will compare the average attendance when each promotion is present vs that when it is absent. This should help identify which if any promotions are the most effective.

```
In [6]: # Select the promotional columns
promo_cols = ["cap", "shirt", "fireworks", "bobblehead"]

# Create an empty dictionary to hold the promotional details
promo_attendance = {}

# For each promotion determine the mean attendance and store in the dictionary
for promo in promo_cols:
```

```

promo_attendance[promo] = df.groupby(promo)["attend"].mean()

promo_attendance_df = pd.DataFrame(promo_attendance)
promo_attendance_df

```

```

Out[6]:

```

	cap	shirt	fireworks	bobblehead
NO	41112.240506	40824.551282	41032.179104	39137.928571
YES	38189.500000	46643.666667	41077.857143	53144.636364

Based on the displayed information it would appear that bobbleheads have the greatest impact and fireworks the least based on quick comparison between NO (promotion not available) and YES (promotion available) mean values. One thing of note is these values do not compare only to days where no promotions are present, only days when that specific promotion is not available. It is also possible more detailed review for combinations of promotion could lead to increased attendance impact.

To make the promotion information easier to interpret and visualize the above comparisons boxplots will be created for each promotion. This will allow me to see the averages but also variability and possible outliers.

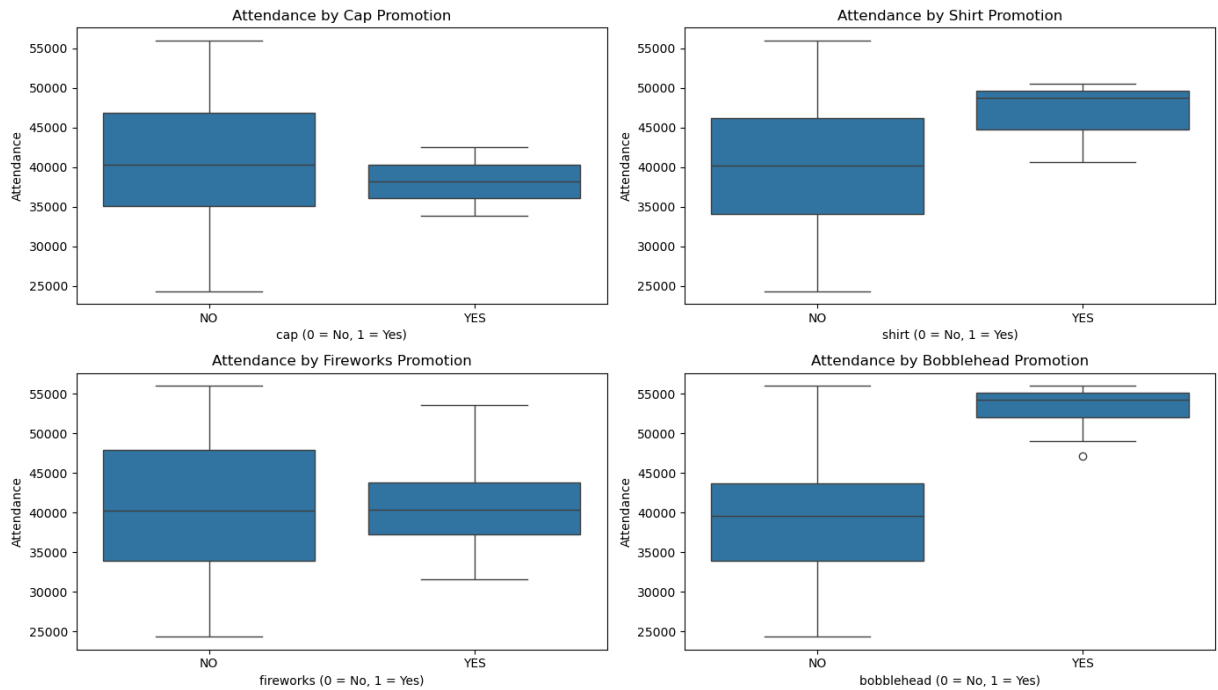
```

In [7]: plt.figure(figsize=(14,8))

for i, promo in enumerate(promo_cols, 1):
    plt.subplot(2, 2, i)
    sns.boxplot(
        data=df,
        x=promo,
        y="attend"
    )
    plt.title(f"Attendance by {promo.capitalize()} Promotion")
    plt.xlabel(f"{promo} (0 = No, 1 = Yes)")
    plt.ylabel("Attendance")

plt.tight_layout()
plt.show()

```



Based on the boxplots, it becomes clear bobbleheads relate to the strongest attendance and has low variance compared to the other promotions. It also shows the cap promotions while having less overall variance then when there is no promotion is actually associated with a lower average attendance. For presentation to management I would not want to use boxplots but bar charts to emphasize the potential gain the promotions offer to attendance.

```
In [9]: # Create an empty dictionary for the promotional effect calculations
promo_effect = {}

# For the calculated means find the difference between when the promotion is present
for promo in promo_cols:
    means = df.groupby(promo)["attend"].mean()
    promo_effect[promo] = means[1] - means[0]

promo_effect_df = (
    pd.DataFrame.from_dict(promo_effect, orient="index", columns=["Attendance Lift"])
    .sort_values(by="Attendance Lift", ascending=False)
)

print(promo_effect_df)

# Create bar chart
plt.figure(figsize=(8,5))
sns.barplot(
    x=promo_effect_df.index,
    y=promo_effect_df["Attendance Lift"]
)

plt.title("Attendance Lift by Promotion Type")
plt.xlabel("Promotion")
plt.ylabel("Increase in Average Attendance")
plt.show()
```

	Attendance Lift
bobblehead	14006.707792
shirt	5819.115385
fireworks	45.678038
cap	-2922.740506

C:\Users\Laurie\AppData\Local\Temp\ipykernel_35312\2237726577.py:7: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame behavior). To access a value by position, use `ser.iloc[pos]`

```
promo_effect[promo] = means[1] - means[0]
```

C:\Users\Laurie\AppData\Local\Temp\ipykernel_35312\2237726577.py:7: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame behavior). To access a value by position, use `ser.iloc[pos]`

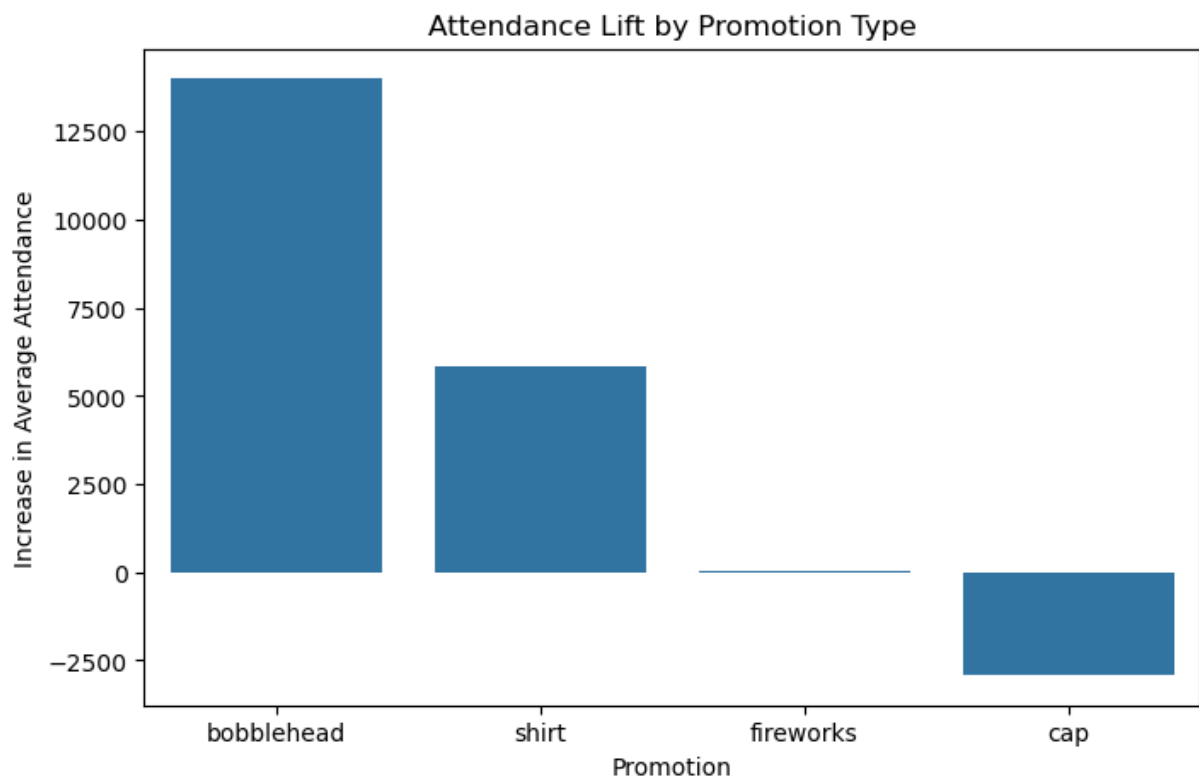
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promo_effect[promo] = means[1] - means[0]
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```
promo_effect[promo] = means[1] - means[0]
```



This creates a very clear picture of the difference in the different promotions and the average differences that will really drive home the pitch to management.

Data-Driven Recommendation to Improve Game Attendance

Based on exploratory data analysis of the 2022 Dodgers attendance data, the below summarizes key findings and provides a data-driven recommendation to improve game attendance through targeted promotion strategies.

Summary of Analysis

The analysis focused on evaluating the attendance patterns by the day of the week and the impact of promotional campaigns, specifically caps, shirts, fireworks, and bobbleheads. Average attendance was calculated across days and compared between game with and without each promotion to identify which strategies were the most effective at driving turnout.

It was identified that Monday and Wednesday games consistently had lower average attendance relative to other days of the week. These days represent the greatest opportunity for improvement leveraging promotional strategies. These could lead to incremental increases that would yield meaningful gains without competing against already high demand games. Among the promotional variables analyzed bobblehead giveaways produced the highest attendance lift. Games with bobblehead promotions showed a clear and substantial increase in average attendance compared to games without the promotion. The other promotions shows more modest or inconsistent results.

Recommendation

Based on these findings, I am recommending that we prioritize bobblehead giveaway promotions as the primary promotion strategy given their demonstrated ability to significantly increase attendance. These promotions should be targeted for Monday and Wednesday games, as they represent the lowest baseline attendance and therefore have the highest potential return on the promotion investment. Marketing and communication efforts would need to align around these targeted promotional games to maximize awareness and impact.

This focused approach allows the organization to deploy promotional resources more efficiently while addressing specific segments of the schedule where attendance were underperforming and gains are more achievable.